

**Gurugram University Scheme of Studies and Examination**  
**Bachelor of Technology (Electronics & Communication Engineering)**

**Semester VIII**

| S. No        | Category | Course Code | Course Title                      | Hours per week |   |    | Credits   | Marks for Sessional | Marks for End Term Examination | Total      |
|--------------|----------|-------------|-----------------------------------|----------------|---|----|-----------|---------------------|--------------------------------|------------|
|              |          |             |                                   | L              | T | P  |           |                     |                                |            |
| 1            | ESC      |             | MOOC –I (Essential)               | 3              | 0 | 0  | 3         | 25                  | 75                             | 100        |
| 2            | ESC      |             | MOOC-II (Essential)               | 3              | 0 | 0  | 3         | 25                  | 75                             | 100        |
| 3            | PROJECT  |             | Project – III/Industrial Training | 0              | 0 | 16 | 8         | 150                 | 150                            | 300        |
| <b>Total</b> |          |             |                                   | <b>22</b>      |   |    | <b>14</b> | <b>200</b>          | <b>300</b>                     | <b>500</b> |

**NOTE: At the end of the 8th semester, each student has to submit the certificate of MOOCs (Essential).**

### MOOC-1(ESSENTIAL)

|                  |                            |   |   |         |                |  |
|------------------|----------------------------|---|---|---------|----------------|--|
| Course Code      |                            |   |   |         |                |  |
| Category         | Engineering Science Course |   |   |         |                |  |
| Course title     | MOOC-1 (Essential)         |   |   |         |                |  |
| Scheme           | L                          | T | P | Credits | Semester: VIII |  |
|                  | 3                          | 0 | 0 | 3       |                |  |
| Class Work       | 25 Marks                   |   |   |         |                |  |
| Exam             | 75 Marks                   |   |   |         |                |  |
| Total            | 100 Marks                  |   |   |         |                |  |
| Duration of Exam | 3Hrs                       |   |   |         |                |  |

A student has to complete NPTEL/SWAYAM Courses of 12 Weeks respectively through MOOCs. For registration to MOOCs Courses, the students shall follow NPTEL/SWAYAM Site <http://nptel.ac.in/> as per the NPTEL policy and norms. The students can register for these courses through NPTEL/SWAYAM directly as per the course offering in Odd/Even Semesters at NPTEL/SWAYAM. These NPTEL/SWAYAM courses (recommended by the University) may be cleared during the B. Tech degree program (not necessary one course in each semester). After successful completion of these MOOCs courses the students, shall, provide their successful completion NPTEL/SWAYAM status/certificates to the University (COE) through their college of study only.

## MOOC-2(ESSENTIAL)

|                  |                            |   |   |         |                |  |
|------------------|----------------------------|---|---|---------|----------------|--|
| Course Code      |                            |   |   |         |                |  |
| Category         | Engineering Science Course |   |   |         |                |  |
| Course title     | MOOC-2 (Essential)         |   |   |         |                |  |
| Scheme           | L                          | T | P | Credits | Semester: VIII |  |
|                  | 3                          | 0 | 0 | 3       |                |  |
| Class Work       | 25 Marks                   |   |   |         |                |  |
| Exam             | 75 Marks                   |   |   |         |                |  |
| Total            | 100 Marks                  |   |   |         |                |  |
| Duration of Exam | 3Hrs                       |   |   |         |                |  |

A student has to complete NPTEL/SWAYAM Courses of 12 Weeks respectively through MOOCs. For registration to MOOCs Courses, the students shall follow NPTEL/SWAYAM Site <http://nptel.ac.in/> as per the NPTEL policy and norms. The students can register for these courses through NPTEL/SWAYAM directly as per the course offering in Odd/Even Semesters at NPTEL/SWAYAM. These NPTEL/SWAYAM courses (recommended by the University) may be cleared during the B. Tech degree program (not necessary one course in each semester). After successful completion of these MOOCs courses the students, shall, provide their successful completion NPTEL/SWAYAM status/certificates to the University (COE) through their college of study only.

## INDUSTRIAL PROJECT/PROJECT-III

|                  |                                |   |    |         |                |  |
|------------------|--------------------------------|---|----|---------|----------------|--|
| Course Code      |                                |   |    |         |                |  |
| Category         | Project                        |   |    |         |                |  |
| Course title     | Industrial Project/Project-III |   |    |         |                |  |
| Scheme           | L                              | T | P  | Credits | Semester: VIII |  |
|                  | 0                              | 0 | 16 | 8       |                |  |
| Class Work       | 150 Marks                      |   |    |         |                |  |
| Exam             | 150 Marks                      |   |    |         |                |  |
| Total            | 300 Marks                      |   |    |         |                |  |
| Duration of Exam | 3Hrs                           |   |    |         |                |  |

### Course objectives:

1. To allow students to demonstrate a wide range of the skills learned during their course of study by asking them to deliver a product that has passed through the design, analysis, testing and evaluation
2. To encourage multidisciplinary research through the integration learned in a number of courses.
3. To allow students to develop problem solving, analysis, synthesis and evaluation skills.
4. To encourage teamwork.
5. To improve students' communication skills by asking them to produce both a professional report and a professional poster and to give an oral presentation

The students are required to undergo industrial training or institutional project work of duration not less than 4 months in a reputed organization or concerned institute. The students who wish to undergo industrial training, the industry chosen for undergoing the training should be at least a private limited company. The students shall submit and present the midterm progress report at the institute. the presentation will be attended by a committee. alternately the teacher may visit the industry to get the feedback of the student.

The final Viva voice of the industrial training or institutional project work will be conducted by an external examiner and one external examiner appointed by the institute. External examiner will be from the panel of examiners submitted by the concerned institute approved by the board of studies in engineering and technology. Assessment of industrial training or institutional project work will be based on seminar, viva-voice, report and certificate of industrial training or institutional project work obtained by the student from the industry or institute.

The internal marks distribution for the students who have undergone industrial training consist of 150 marks internally and 150 marks by an external examiner.

### Course outcomes

On successful completion of the course students will be able to:

1. Demonstrate a sound technical knowledge of their selected project topic.
2. Undertake problem identification, formulation and solution.
3. Design engineering solutions to complex problems utilising a systems approach.
4. Conduct an engineering project.
5. Communicate with engineers and the community at large in written and oral form.
6. Demonstrate the knowledge, skills and attitudes of a professional engineer.